

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A mixture consisting of only vitamin D and calcium has a total mass of **150** grams. The mass of vitamin D in the mixture is **50** grams. What is the mass, in grams, of calcium in the mixture?

Answer

- A. **200**
- B. **150**
- C. **100**
- D. **50**

Correct Answer: C

Rationale

Choice C is correct. Let *d* represent the mass, in grams, of vitamin D in the mixture, and let *c* represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is **150** grams. Therefore, the equation *d* + *c* = **150** represents this situation. It's also given that the mass of vitamin D in the mixture is **50** grams. Substituting **50** for *d* in the equation *d* + *c* = **150** yields **50** + *c* = **150**. Subtracting **50** from both sides of this equation yields *c* = **100**. Therefore, the mass of calcium in the mixture is **100** grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.